



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECD2613 – ANALISIS DAN REKABENTUK SISTEM (WBL)

(SYSTEM ANALYSIS AND DESIGN) (WBL)

SECTION 01

ASSIGNMENT 1

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Question 1

- a. Technical feasibility, operational feasibility and economic feasibility
- b. The operational feasibility of the proposed project is whether the new system can work as intended and whether the people will still use the new system.
- c. The economic feasibility of the proposed project is the total financial costs to analyse the new system requirements and the costs of building a new hardware or software system.
- d. The technological feasibility of the proposed project is whether the new technology in the enhanced system will improve the user's experience and whether the new function will disrupt the original function in the existing system.
- e. Based on the conversation between Ahmad and Abu, I would still recommend they do a full-blown systems study. Even if the conversation really shows that Abu's rejection of Ahmad's proposal is justified, Ahmad's view, which is looking far ahead into the market and always seeking advancement in their current system is not always a bad thing. As mentioned before it is still technologically feasible, there are definitely new improvements and innovations they can implement in their current system to enhance their company's reputation and sales. All they need is a better vision of the proposed project's business goals, assessing the potential risks and identifying alternative methods to attract new types of consumers and rank their company even higher in the food distributing market.

Question 2

Task Number	Task Description	Duration	Predecessor
1.0	Research Phase		
1.1	Define the scope and objective of the project	0.5 weeks	-
1.2	Determine the project team	0.5 weeks	-
1.3	Project team meeting	0.5 weeks	-
1.4	Having an interview with the customer to collect the requirement	0.5 weeks	1.1
2.0	Planning		
2.1	Gather, analyse and documentation requirements from the customer	1 weeks	1.4
2.2	Investigate other system	1 weeks	1.5
2.3	Planning budget	0.5 weeks	2.1
2.4	Prepare proposal	1 weeks	2.3,2.1
2.5	Get approval for the financial budget	0.5 weeks	2.3
2.6	Introduce prototype to customer	1 weeks	2.4
3.0	Design		
3.1	Write the system design report	0.5 weeks	2.1
3.2	Design system architecture	1 weeks	3.1
3.3	Design user interface	1 weeks	3.1
3.4	Design database	1 weeks	3.2
4.0	Development		
4.1	Develop database	1 weeks	3.3
4.2	Develop backend	1 weeks	4.1
4.3	Develop frontend	1 weeks	4.2
4.4	Implement a new computer system	0.5 weeks	4.1,4.2,4.3
4.5	Documenting the system	Throughout the project	3.1
5.0	Testing		
5.1	Conduct system testing	1weeks	4.4
5.2	Do the user training	1weeks	5.1
6.0	Evaluation		
6.1	Collect feedback from the user	1 weeks	5.1
6.2	Maintaining and improving the system	1 weeks	6.1
6.3	Monitor the progress of the task	ongoing	-

Question 3

Rather than relying on tightly scheduled meetings, the agile method emphasises daily informal communication through messaging platforms or virtual communication platforms from time to time for greater collaboration and improvisation.

The second is a faster decision making system as the traditional method usually follows the hierarchical structure where higher ranking personnels make the final decisions, whereas with agile method it encourages discussions between team members no matter their rank as the authority is distributed equally so everyone can always give their feedback at any time.

The third is the improvement in adaptability and flexibility of team members. This is because agile methods emphasise adaptability to changing system requirements while traditional project management approaches often follow a fixed plan or roadmap. This would ultimately reduce the risk of financial cost, leading to potential cost savings and increased efficiency.

Besides, the agile method targets to deliver working increments of the product within a short time. This reduces cycle time and allows for faster feedback loops, unlike traditional projects which typically follow a sequential approach. This can save a lot of time in the long run.

Lastly, the traditional method limits each team member to their own respective professions and responsibilities, causing weak chemistry between team members in the company. Agile methods solve this by encouraging team members to work collaboratively constantly, fostering a sense of shared responsibility and commitment to make the project a success.

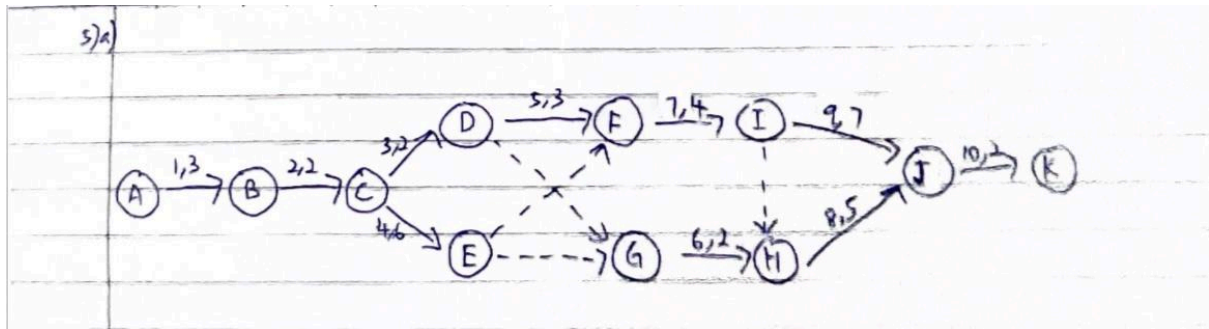
Question 4

Costs	Year 0	Year 1	Year 2	Year 3
Development Costs				
Hardware	50000			
Software	15000			
Training	30000			
Consulting	80000			
Data Conversion	40000			
Total	215000			
Production Costs				
Supplies		15000	15000	15000
IS Salaries		75000	82500	90750
Upgrades		10000	10000	10000
Annual Production Costs (Total)		100000	107500	115750
Annual Production Costs (Present Value)		74074	58985	47046
Accumulated Costs		289074	348059	395105
Benefits	Year 0	Year 1	Year 2	Year 3
Improve Customer Service		150000	180000	216000
Increase productivity		150000	187500	234375
Annual Benefits(Total)		300000	367500	450375
Present Value		222222	201646	183051
Accumulated benefits (Present Value)		222222	423868	606919
Gain or Loss		-66852	75809	211814
Profitability Index	211814/ 215000 = 0.99			

The profitability index (PI) value for this PV analysis is 0.99, showing that the SKORBISTARI project proposal is not a good investment because the PI is less than one.

Question 5

5a)



5b) Path 1: A-B-C-D-F-I-J-K

Path 2: A-B-C-E-G-H-J-K

Path 3: A-B-C-D-G-H-J-K

Path 4: A-B-C-E-F-I-J-K

Path 5: A-B-C-D-F-I-H-J-K

Path 6: A-B-C-E-F-I-H-J-K

5c) Path 1: A-B-C-D-F-I-J-K (23 WEEKS)

Path 2: A-B-C-E-G-H-J-K (20 WEEKS)

Path 3: A-B-C-D-G-H-J-K (16 WEEKS)

Path 4: A-B-C-E-F-I-J-K (24 WEEKS)

Path 5: A-B-C-D-F-I-H-J-K (21 WEEKS)

Path 6: A-B-C-E-F-I-H-J-K (22 WEEKS)

Path 4 is critical path which is A-B-C-E-F-I-J-K that is **24**.

5d)

[illegible]